

## PROGRAMMING ASSIGNMENT - 3

| Question No. | Line No. | Error Type       | Error                                                                       | Error Correction                                                |
|--------------|----------|------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------|
| problem1.c   | 14       | Syntax Error     | Semicolon missing at the end of statement.                                  | Add ; at line end:<br><b>int total = 0;</b>                     |
| problem2.c   | 28       | Memory Error     | malloc used but free() function not called before return.                   | Add <b>free(node);</b> before return statement.                 |
| problem3.c   | 42       | Logic Error      | Loop going out of index due to using <= instead of <.                       | Change condition:<br><b>for(i=0;i&lt;MAX; i++)</b>              |
| problem4.c   | 19       | Warning          | Wrong format specifier %d used for float value.                             | Use %f instead of %d inside printf.                             |
| problem5.c   | 8        | Runtime Error    | Forgot to put & symbol in scanf statement.                                  | Change to:<br><b>scanf("%d", &amp;val);</b>                     |
| problem6.c   | 35       | Null Pointer     | File opened with fopen but checked directly without checking if it is NULL. | Add condition:<br><b>if (fp == NULL)<br/>return;</b>            |
| problem7.c   | 51       | Garbage Value    | Local variable used in loop without initializing it with 0.                 | Initialize at top:<br><b>intsum = 0;</b>                        |
| problem8.c   | 22       | Security Warning | Using unsafe gets() function for string input.                              | Replace with safe function:<br><b>fgets(str, 100, stdin);</b>   |
| problem9.c   | 12       | Compiler Error   | Standard library header file not included for malloc.                       | Include header file at top:<br><b>#include &lt;stdlib.h&gt;</b> |
| problem10.c  | 64       | Logic Error      | Used single equals = instead of == in if statement condition.               | Change to:<br><b>if (x == 10)</b>                               |